TNT Video Kahoot: answer key

Covers: ERP video, DIY Ganglion BCI video, oscilloscope video, motor imagery demo, and the Riemannian explainer.

1. ERP stands for...
Event-related potential (correct)
Electrical resting pulse
Eye reflex pattern
Evoked rhythm power
Why: A small voltage wave the brain makes at a fixed time after an event.

2. In "P300", what does the P mean?
Power
Positive peak (correct)
Pulse
Parietal
Why: Peaks are named P (positive) or N (negative).

3. In "P300", what does the 300 mean?
About 300 ms after the event (correct)
300 microvolts
300 channels
300 trials
Why: The number is roughly when the peak shows up.

4. Why do ERP studies average many trials?
It makes files smaller
The brain gets louder
Random noise cancels, the ERP adds up (correct)
To dry the electrodes
Why: The ERP happens at the same time every trial; background EEG doesn't.

5. An N200 is...
A 200 Hz rhythm
A negative dip ~200 ms after an event (correct)
200 neurons firing
A noise level
Why: N = negative, 200 = about 200 ms.

6. How many EEG channels does the OpenBCI Ganglion have?
2
8
16
4 (correct)
Why: 4 channels. The NeuroPawn Knight Board has 8.

7. What does BrainFlow do?
Builds the headset
Streams data from the board into Python (correct)
Cleans the electrodes
Only trains EEGNet
Why: It's the bridge between the hardware and your code.

8. In the DIY BCI video, what makes the motor move?

The signal crosses a threshold (correct)
A voice command
A key press
A timer
Why: Simple rule: brain signal goes above a set level, motor turns on.

9. A jaw clench could also cross that threshold. That's called...
A P300
Alpha
An artifact (correct)
A covariance
Why: Muscle signals are much bigger than EEG. That's our research question.

10. Closing your eyes boosts which rhythm at the back of the head?
Beta (13-30 Hz)
Delta (1-4 Hz)
Gamma (30+ Hz)
Alpha (8-12 Hz) (correct)
Why: Shown in the oscilloscope video and in our own Ganglion demo.

11. In the motor imagery demo, what moves the circle?
Eye blinks
Imagined hand and foot movements (correct)
Real arm movements
Voice
Why: Imagining a movement changes the mu rhythm over motor cortex.

12. A Riemannian classifier mostly looks at...
How channels vary and move together (correct)
Video of the face
Heart rate
One channel's raw voltage
Why: Each trial becomes a covariance matrix: power per channel + how channels relate.

13. Why can't the Riemannian method work with just 1 channel?
It's too slow
It needs 64 channels
It only works on MEG
No between-channel info to use (correct)
Why: With one channel there's nothing to relate it to, so we use band power + LDA instead.